DS5500 Data Science Capstone Project Proposal

1. Project Title

Personalized Skincare Routine Recommendation System Using Ingredient-Aware Machine Learning

2. Problem Statement

Choosing an effective skincare routine is challenging for many consumers due to the large number of available products, varying ingredient formulations, and individual differences in skin type, concerns, and budget constraints. Most users rely on trial-and-error, influencer recommendations, or generic product filters, which often fail to account for ingredient compatibility, skin sensitivity, or overall routine cost.

This problem is important because improper product selection or incompatible routines can lead to skin irritation, wasted spending, and poor user outcomes. In real-world settings, consumers need guidance that balances personalization, safety, and affordability rather than recommending isolated products. Existing skincare recommendation tools often focus on individual products or popularity metrics rather than constructing safe, compatible routines tailored to individual constraints.

The goal of this project is to build a personalized skincare routine recommendation system that generates complete AM and PM routines based on user-specific inputs such as skin concerns, skin type, active tolerance, preferences, and total budget constraints. The system leverages structured product data and ingredient-level features to recommend compatible and explainable routines rather than single products.

Our team selected this problem because it represents a practical, consumer-facing application of recommender systems and machine learning, while allowing us to work with real-world, messy data and focus on explainability, constraint-aware recommendation, and user-centered design.

3. Team Structure, Contributions, and Objectives

Team Objectives

- Explore, clean, and preprocess real-world skincare product datasets.
- Engineer meaningful product-level and ingredient-level features to support personalized recommendations.
- Design and implement a constraint-aware recommendation pipeline that accounts for user preferences, skin characteristics, and budget limits.
- Build an interactive user interface to demonstrate the system end-to-end.
- Collaborate effectively using GitHub for version control, documentation, and code review.

Team Member Contributions

Yashi:
Problem formulation and overall recommendation system design, including defining user inputs, personalization constraints, and objectives. Dataset exploration and initial data cleaning, along with planning and experimentation of different recommendation approaches (such as content-based and constraint-aware methods) to identify effective strategies for routine-level recommendations.

Anjali:
Exploratory data analysis (EDA) and dataset understanding, including analysis of product categories, ingredient distributions, and pricing trends. Feature understanding at the product and ingredient level, preliminary visualizations, and support for model experimentation and evaluation to inform feature selection and system design.

Roshita:
User experience planning and front-end interface design to showcase personalized AM and PM skincare routines. Integration of model outputs into the interface, end-to-end system testing, and responsibility for repository organization, GitHub workflow practices, and project documentation.

Individual Learning Objectives

Yashi:

- Develop hands-on experience designing and experimenting with recommender system algorithms, including constraint-aware and content-based approaches.
- Strengthen skills in data preprocessing and feature engineering for real-world, unstructured and semi-structured datasets using Python-based tools.

Anjali:

- Improve exploratory data analysis and visualization skills to extract insights from complex product and ingredient datasets.
- Learn to support model evaluation and experimentation by comparing recommendation strategies and understanding trade-offs between different modeling choices.

Roshita:

- Gain experience building interactive user interfaces and dashboards using frameworks such as Streamlit or Flask to present model outputs effectively.
- Develop skills in integrating machine learning outputs into end-to-end applications, including version control, documentation, and collaborative workflows.

4. GitHub Setup and Collaboration

All team members will collaborate using a shared GitHub repository created specifically for this project, following a clear and descriptive naming convention that reflects the skincare recommendation system topic.

A feature-branch workflow will be used throughout the iteration. Each team member will work on separate branches for tasks such as data exploration, modeling, or interface planning, and will submit pull requests before merging changes into the main branch.

Pull requests will be reviewed by at least one other team member to ensure code quality, consistency, and alignment with project goals.

Commits will be frequent, incremental, and clearly documented, with descriptive commit messages that summarize the changes made. This approach ensures transparency and enables effective collaboration and version control.

Role assignments will include a repository owner responsible for managing the main branch and resolving merge conflicts, while other team members will act as reviewers for pull requests during this iteration.

The README file will serve as the central documentation hub, describing the project scope and objectives, datasets being used, tools and libraries, and setup instructions.

5. Tools and Programming Languages

Programming Languages

- **Python** – primary language for data processing, model development, and system integration.

Libraries and Frameworks

- **Pandas, NumPy** – preprocessing skincare product datasets and engineering ingredient-derived features.
- **PyTorch** – training neural network models for user and product embeddings.
- **Sentence-Transformers / Hugging Face Transformers** – building dual-encoder architectures for neural retrieval.
- **Scikit-learn** – baseline retrieval methods, similarity metrics, and evaluation.

Tools and Platforms

- **Jupyter Notebook** – data exploration, feature validation, and model experimentation.
- **GitHub** – collaborative development, version control, and project documentation.
- **Kaggle** – acquisition of skincare product and ingredient datasets.
- **Streamlit** – development of an interactive interface to collect user inputs and display recommended routines.
- **FAISS or similar vector search libraries** – efficient similarity search over product embeddings.

This project primarily strengthens existing Python and data analysis skills while introducing new experience in recommender systems, constraint-aware modeling, and interactive application development.

6. Cluster Access Request

Cluster access required: No
Explanation: This project will be developed using moderate-sized datasets and standard machine learning workflows that can be handled on local machines or cloud notebook environments. No high-performance computing or cluster resources are required for this iteration.